High-Level Design (HLD)

Project: Aarvii CCTV AMC Management System **Phase:** D0 — Project Foundation Documentation **Source of truth:** requirements-freeze-v1.md · Platform baseline: platform-discovery-report.md

> Design-level document. No database design or ER diagrams here (deferred to D0-4). No code.

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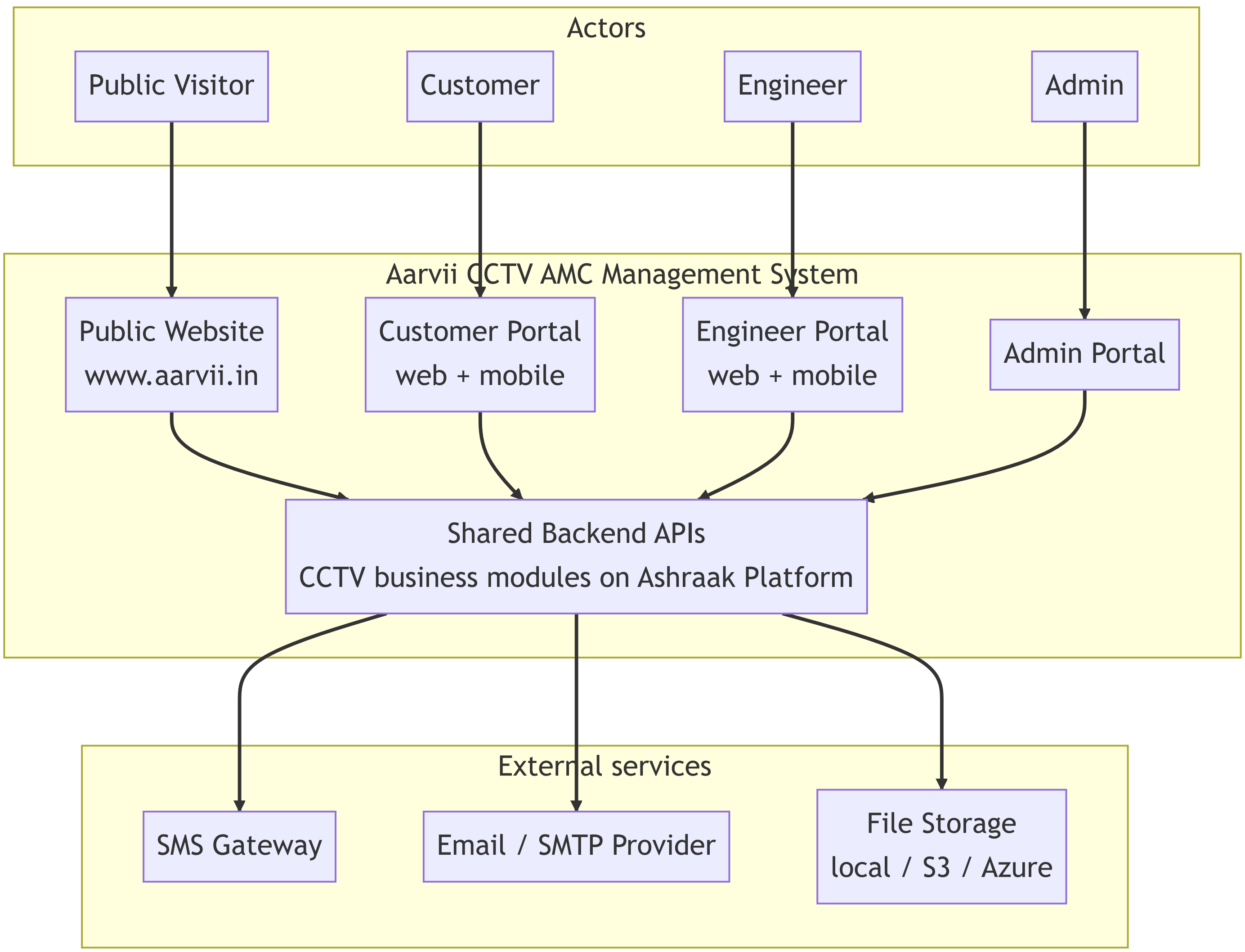
1. System context



The four business applications **share a common backend** (freeze §2). The backend is the frozen **Ashraak Platform V1** plus new **CCTV business modules** (freeze §20 mandates reuse, no duplicate implementation).

2. Architecture overview

Key principle: CCTV business modules consume Core capabilities **only through published contracts** (`IFileStorage` , `INotificationService` , `IAuditService` , `IWebhookPublisher` , auth/tenant contracts). Core is never modified (business-module-policy).

3. Applications (freeze §2)

Application	Channel	Users	Purpose
Public Website	Web (www.aarvii.in)	Public Visitor	Showcase, AMC plan info, lead generation (Get Quote, AMC Inquiry), login entry
Customer Portal	Web + Flutter app	Customer	Dashboard, AMC details, service history, upcoming visits, tickets, invoices, profile, password reset
Engineer Portal	Web + Flutter app	Engineer	Assigned visits/tickets, visit reporting, photo upload, GPS, selfie, signature, ticket creation
Admin Portal	Web	Admin	All 11 management areas: leads, customers, sites, assets, AMC plans, contracts, scheduling, tickets, engineers, invoices, reporting

Detail: application-architecture.md

4. Modules (freeze §4)

15 approved modules; responsibilities and boundaries in module-architecture.md:

Public Website · Lead Management · Customer Management · Site Management · Asset Management · AMC Plans · AMC Contracts · Service Scheduling · Visit Management · Ticket Management · Engineer Management · Invoice Management · Reporting · Customer Portal · Engineer Portal

5. Integrations

Integration	Purpose	Freeze ref	Approach
Email provider (SMTP)	Email notifications	§17	Platform `INotificationService` provider model
SMS gateway	SMS notifications and OTPs	§17	New provider integration consumed by the CCTV notification flows
File storage (local/S3/Azure)	Photos, videos, selfies, signatures, PDFs	§12, §19	Platform `IFileStorage`
PDF generation	AMC Contract / Visit Report / Invoice PDFs	§19	Server-side PDF rendering within CCTV modules
Webhooks (outbound)	Event delivery to external systems	§20 (reuse)	Platform webhook catalog + `IWebhookPublisher`

Integration	Purpose	Freeze ref	Approach
API Keys	M2M access where needed	§20 (reuse)	Platform ApiKeys module

6. Technology stack (inherited from platform)

Layer	Technology
Backend	.NET 10, ASP.NET Core Minimal APIs, EF Core 9, OpenIddict
Web	React 19, TypeScript, Vite 6, TanStack Query, Zustand, Theme Engine (`platform-ui` , CoreUI default)
Mobile	Flutter (Android/iOS), Riverpod, go_router, OpenAPI-generated SDK
Data	PostgreSQL (per-module schemas), Redis (cache/sessions/rate limits), MongoDB (audit)
Observability	Serilog + Seq, OpenTelemetry, correlation IDs, health probes
CI/CD	GitHub Actions (ci, docs-validation, mobile, android/ios release)

7. Security

Concern	Design
Authentication	Platform Auth (JWT/OAuth2 via OpenIddict); Login OTP & Password Reset OTP events (§17)
Authorization	Platform RBAC/ABAC mapped to the 4 actors (§3); engineer restrictions enforced server-side (§15)
Data ownership	Customers access only their own contracts, sites, visits, tickets, invoices (§3)
Report gating	Visit reports invisible to customers until admin approval (§13)
Evidence integrity	GPS lat/long/timestamp stored; mandatory completion checklist enforced by the API, not just the UI (§12)
File security	Tenant-scoped storage with access control via platform Files (§20)
Audit	Platform Audit observer captures business events (§20)
Transport/API	Platform host middleware: rate limiting, correlation, API-key auth where applicable

8. Deployment assumptions

#	Assumption
1	Single shared backend deployment (modular monolith) serves all four applications (freeze §2).
2	Public Website and SPA portals are served as web frontends against the same API host; www.aarvii.in fronts the public site.
3	PostgreSQL, Redis, MongoDB, and Seq run per the platform's Docker-based environment locally; production equivalents are provisioned at deployment.
4	Mobile apps are distributed via Play Store / App Store using the platform's existing fastlane release pipelines.

#	Assumption
5	SMS and SMTP providers are configured via environment configuration (validated at startup by the platform).
6	CCTV business modules are feature-flag gateable per platform policy for safe rollout.

9. Design constraints

- **No database design / ERD in D0-1..3** — entity modeling begins in D0-4.
- **Core Platform frozen** — all CCTV functionality lives in business modules (Layer 2 registration).
- **Theme rule** — portal UIs render `platform-ui` primitives; direct theme imports are forbidden.
- **Out-of-scope items** (freeze §21) must not appear in the design.

Related documents

- application-architecture.md
- module-architecture.md
- mobile-architecture.md
- workflow-overview.md
